Student Performance Prediction

An End-to-End Machine Learning Web Application

B.TECH · AI & DATA SCIENCE

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The Problem: Early Academic Risk Detection

The Gap

Students & educators lack early, data-driven insight into academic risk

5 Key Inputs

Study Hours · Attendance · Previous Score · Assignment Score · Sleep Hours

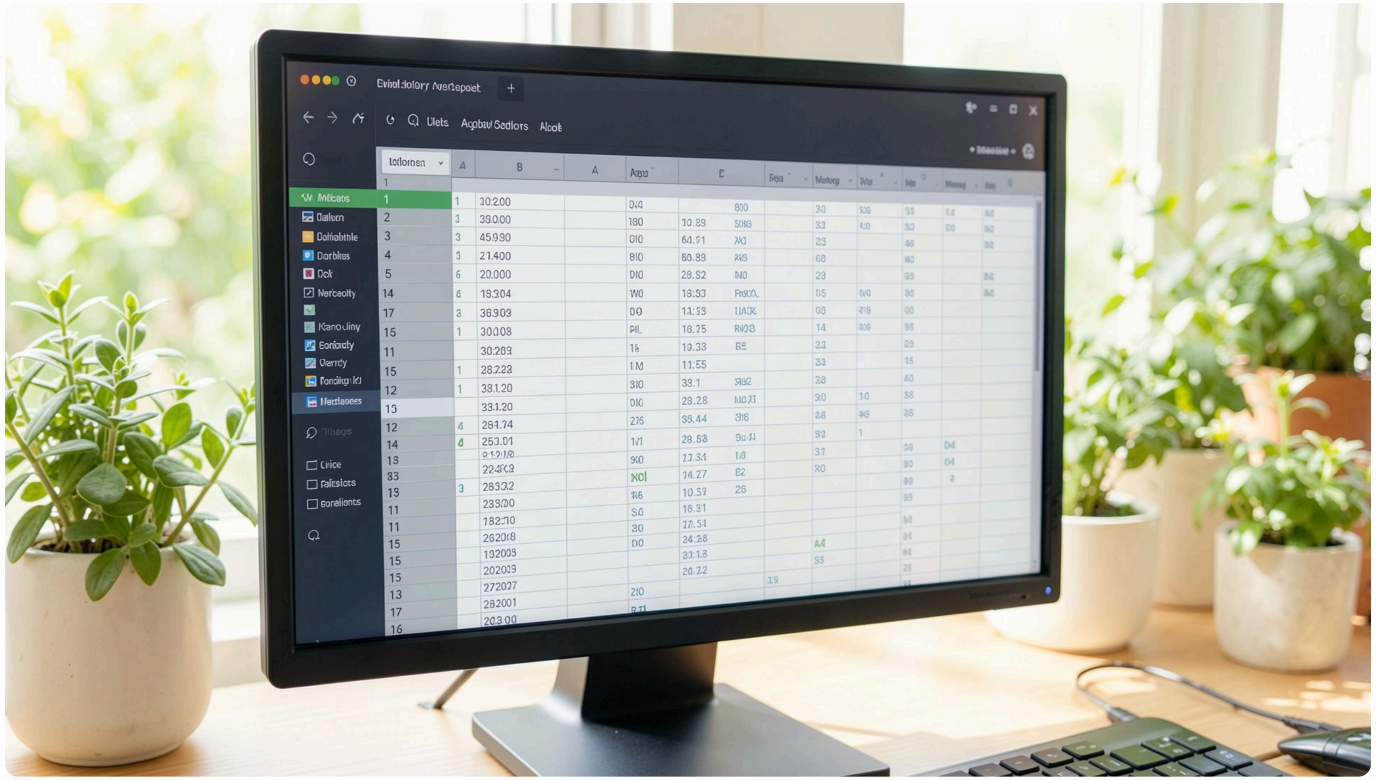
The Goal

Predict final score, analyze factors, give personalized suggestions & track history

Delivered As

Responsive, deployed, multi-user web app — not just a notebook model

Dataset & Data Analysis



DATASET OVERVIEW

300 Records · 6 Columns

Features: Study Hours, Attendance, Previous Score, Assignment Score, Sleep Hours

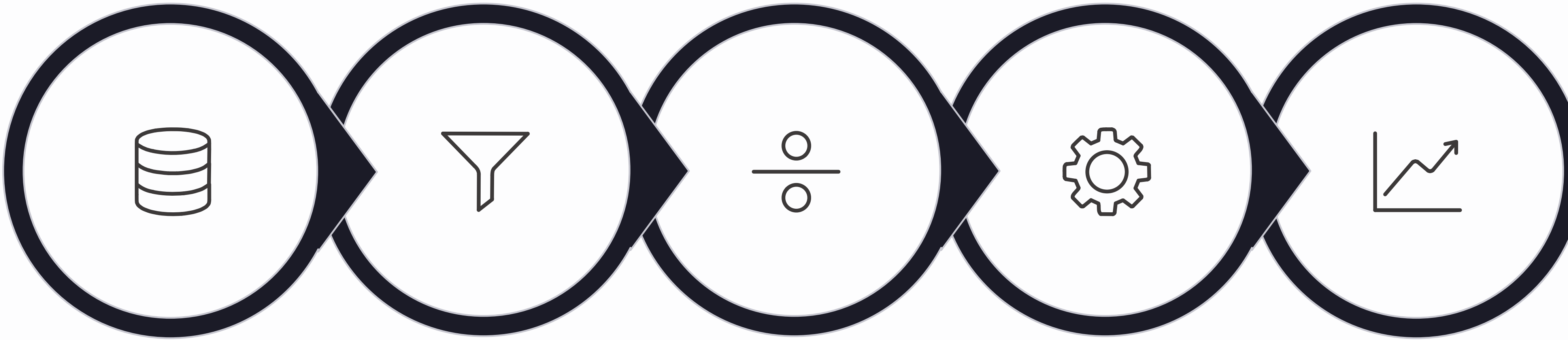
Target: Final Score (out of 100)

PRE-MODELING CHECKS

- Shape, column names, data types
- Missing values & duplicates
- Descriptive statistics
- Feature–target relationships

Result: Clean dataset — zero duplicate records

Machine Learning Approach & Model Selection



Dataset

Preprocessing

Train/Test Split

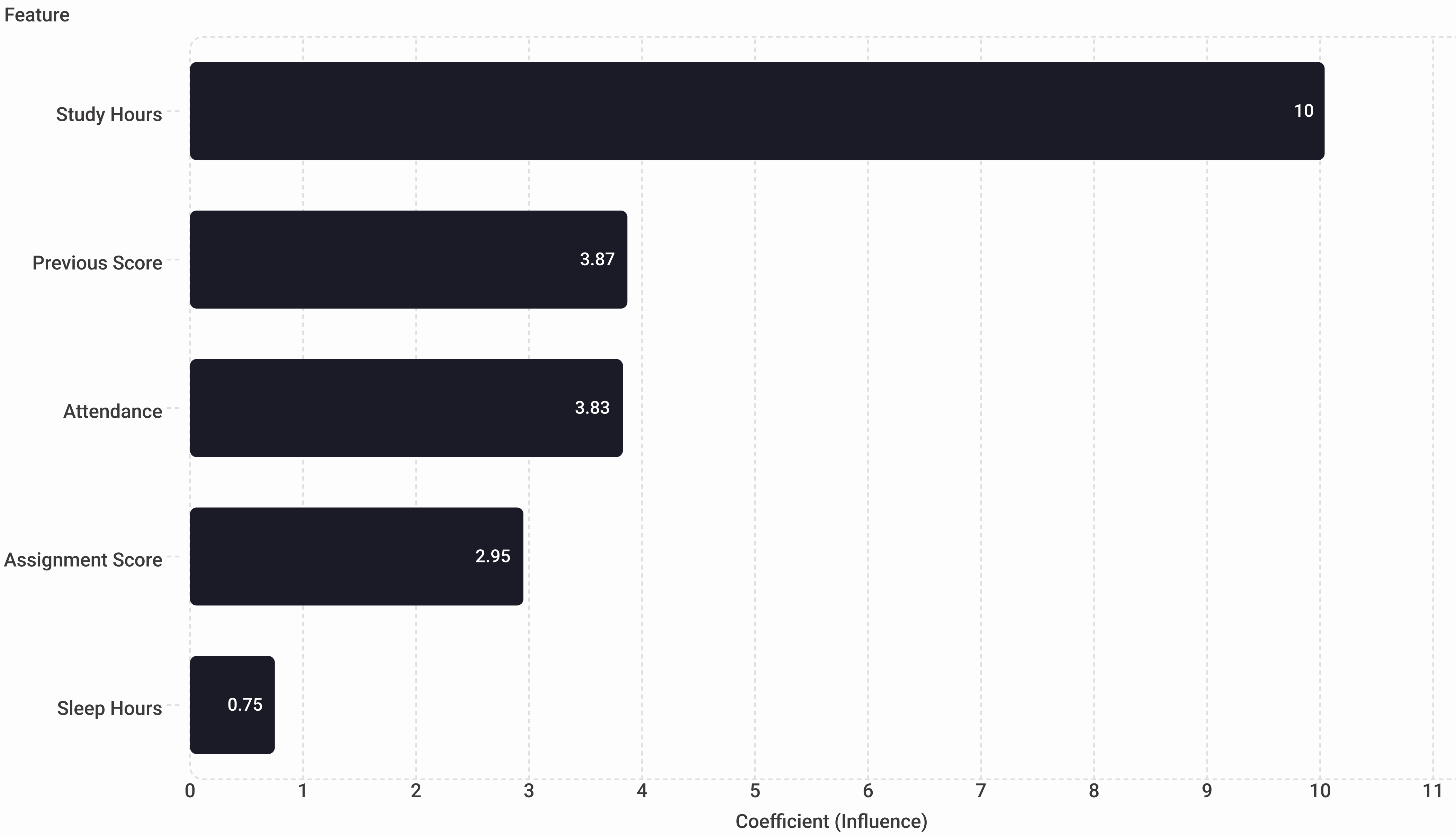
Model Training

Evaluation

Linear Regression	Decision Tree	Random Forest
✅ Selected — best generalization	Tested — overfitting risk	Tested — overkill for dataset size

Libraries: Pandas · NumPy · Scikit-learn · Joblib

Model Performance & Feature Importance



MODEL METRICS

0.799

R^2 Score

Explains ~80% of variance in final score

3.92

MAE

Average error of ~4 points out of 100

24.99

MSE

Mean squared error

System Architecture & Tech Stack

Frontend

HTML · CSS · JavaScript

Backend

Python · FastAPI · Uvicorn · Jinja2

ML Layer

Scikit-learn model serialized with Joblib (.pkl)

Database

PostgreSQL + SQLAlchemy

Auth

Password hashing + session middleware

Deployment

Git → GitHub → Render

Core Application Features

🔐 Secure Auth

Registration & login with password hashing + session authentication

🎯 Prediction Engine

5-input form → instant score + level: Excellent / Good / Average / Needs Improvement

📊 Live Analysis

Dynamic performance analysis — live progress bars per input

💡 Suggestions

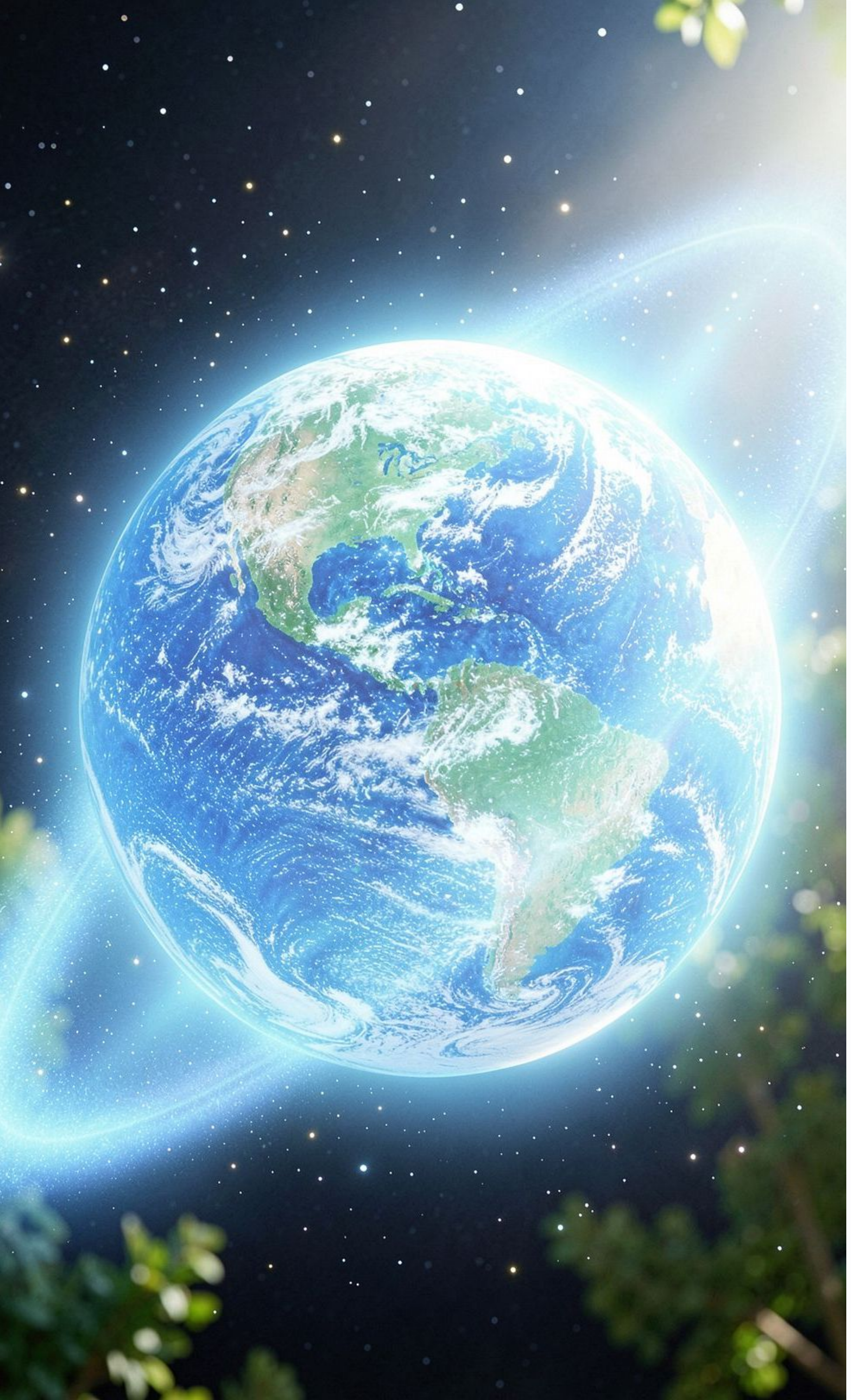
Personalized suggestions engine based on weak input areas

📈 History

Trend graph, best/average score tracking, selective checkbox deletion

👤 Profile

Editable name & DOB; email fixed as account identifier



UI/UX: The Immersive Experience

✨ Glass Design

Glass-style cards, gradient effects, animated score circle & progress bars

🌍 Space Background

Animated Earth/space — stars, glow, orbiting moon, floating particles

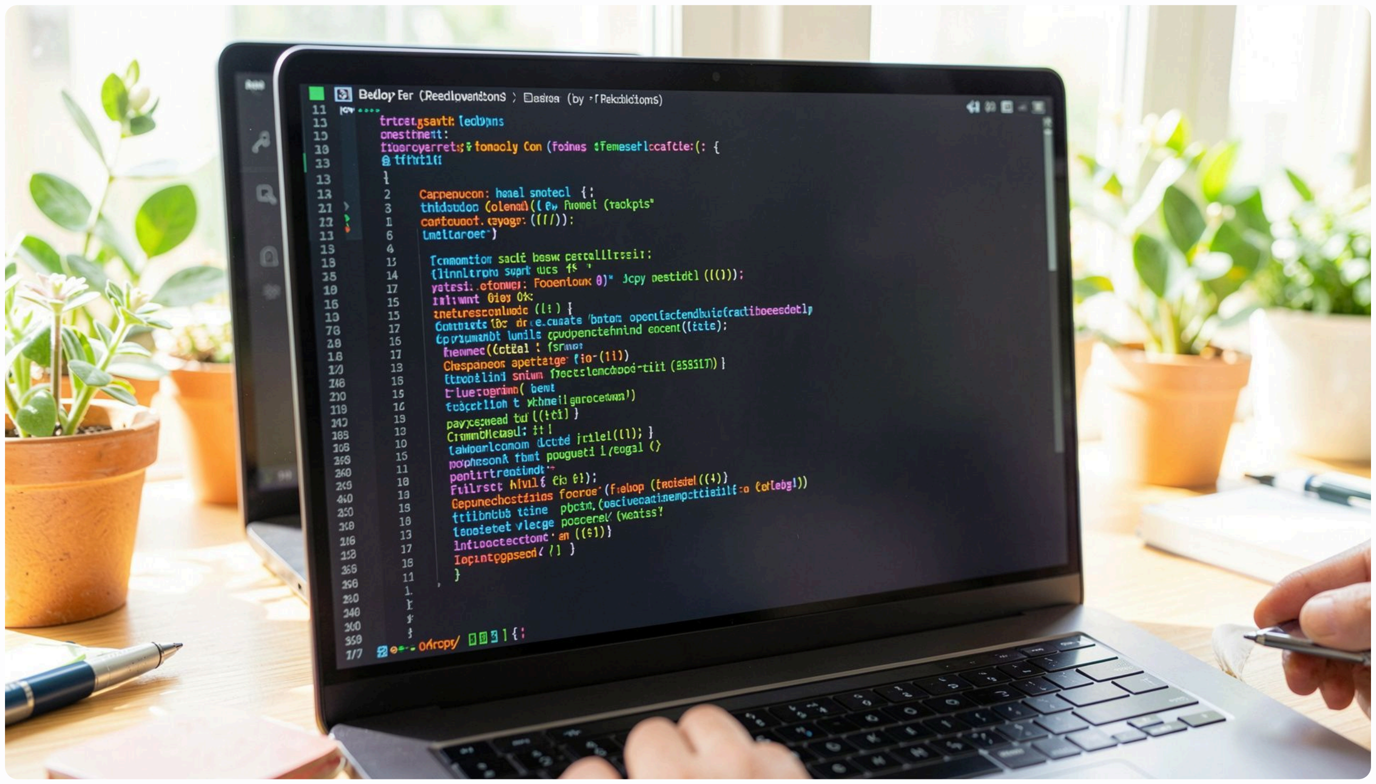
🖱️ Parallax

Mouse-driven parallax — Earth & stars shift subtly with cursor for a 3D feel

📱 Responsive

Fully responsive across laptop and mobile — 25 total features

Engineering Challenges & How They Were Solved



→ psql not recognized

PATH issue → switched to pgAdmin for DB management

→ Render–PostgreSQL failures

Fixed DATABASE_URL & SQLAlchemy connection format

→ relation "users" does not exist

Enforced Base.metadata.create_all() before schema changes

→ Missing packages

python-multipart, Jinja2, auth libs → added to dependency file

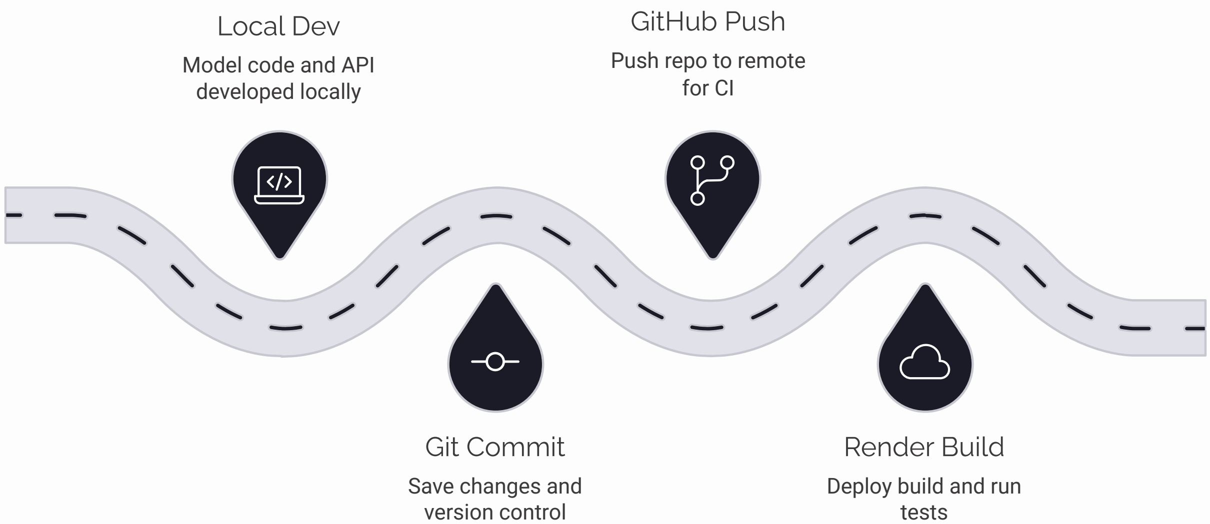
→ 500 error on /login

Debugged via Render logs — traced to DB/auth config

→ Mobile intro video

Resolved cover vs contain cropping + "Tap to Start" for autoplay restriction

Deployment, Learnings & Conclusion



🧠 ML to Production

Full pipeline from training to live prediction

⚡ FastAPI + SQLAlchemy

Backend & database integration mastered

🔐 Auth & Sessions

Real-world authentication implemented end-to-end

🌐 Live Product

ML · backend · frontend · database · deployment — one complete, live product